

In-Class Activities

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These activities are designed to be used in class to help students understand the material. They may or may not be part of the graded work for the class. Please ask your professor if you need help!

If you are reviewing these materials outside of class, you can post to the course Zulip for help, too.

Reading Mathematics

Reading mathematics, and especially mathematics textbooks, is not like reading a novel for pleasure. Often, one sentence of a book or paper contains a paragraph's worth of information. This worksheet will help you develop some habits to read math effectively.

1. **Advice from the Authors.** On page xviii, our textbook authors give advice to a student reader. Summarize the advice here.
2. **Using the End of Section Materials.** Head to the back of Section 1.1 and jot down of the key terms. Leave space to write notes about each term. Do this before you start reading the material in the chapter.

How many terms are in your list?

3. **Reading for Meaning.** The first term in the Key Terms list is **linear equation**. Start reading the chapter and keep an eye out for this term.

We find the first appearance of the term on page 1. It is defined as an equation of the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$ where the variables x_i ($1 \leq i \leq n$) are unknowns and $a_1, \dots, a_n$ and b are real (or complex) numbers.

We can see that it is closely related to the next terms in the Key Terms list, **solution** and **linear system**. Write those definitions in your list of terms, too. It's helpful to include a page number where it appears in the book in case you want to find it again.

4. **Reading Comprehension: Examples and Non-Examples.** When you encounter a definition, it's a good idea to jot down an example and a non-example.

With your team, come up with an example and non-example of a linear system based on page 1 of our textbook. Do not reuse the (non-)examples in the book! Make note of how many equations and how many unknowns are in your examples and non-examples.

Why is your non-example *not* a linear system? Are there other ways to make a non-example?

Continue reading until Example 1 on page 2. Make sure to fill out the definitions of the Key Terms you encounter.

Before you read Example 1, see how the new terms relate to your example of a linear system. Does your linear system have a **solution**? If not, what term should we use to describe it? If so, what would you check next?

5. **Examples in the Textbook.** The authors include an example (Example 1) in the textbook on page 2.

What are the authors trying to demonstrate with the example? That is, what's the point of the example and why is it being shown to the reader now?

Write a note for yourself that "Example 1, page 2, is showing..."

6. **Stamina Time! Keep reading, practicing the habits above..** There is one more key term on page 2; don't miss it

Pages 3-5 contain Examples 2-5. Repeat the previous exercise for each of these examples.

After Example 5, the authors summarize some material and present figures with geometric interpretations. If you have questions about these figures, jot them down so you can ask about them in class (or on Zulip before class).

Finish reading the chapter.

7. **Reading Comprehension: Trying Exercises.** Your professor has assigned some homework problems, but she did not assign number 1. Try that with individually and check with your teammates when you all finish.

Your professor assigned problems 8, 12, and 16. Decide which group member will be the lead recorder for each problem (this role should rotate), and talk through the problems together.

If you are stuck as a team, flag down your professor for help!

Key Terms

linear equation	page 1 an equation like $ax + by = c$ where a, b, c are real (or complex) and x, y are variables. This can have any number of variables and coefficients (not just two).
solution of a linear equation	page
linear system	page
unknowns	page
inconsistent system	page
consistent system	page
homogeneous system	page
trivial solution	page Note: only applies to homogeneous systems
nontrivial solution	page Note: only applies to homogeneous systems
equivalent systems	page
unique solution	page
no solution	page
infinitely many solutions	page
manipulations on linear systems	page
method of elimination	page See also box on page 6!

Examples and Non-Examples

linear system of equations	example non-example
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Group Exercises, Chapter 1, Sections 1-3

We'll review some of the basic concepts from Chapter 1, sections 1-3 in groups, with some exercises to turn in.

Exploring Linear Systems. For each of the following prompts, first try to construct an example (or decide that one cannot exist). Then reflect on what the question reveals about linear systems. Use vocabulary from the chapter in your reflections.

Worksheet Exercise 1 Try These!

- (a) Write down an inconsistent *homogeneous* linear system in two unknowns, or explain why this is impossible.
- (b) Find the solution(s) to the linear system, if any exist.

$$\begin{aligned}x + 4y &= 12 \\ 3x + 8y &= 4\end{aligned}$$

- (c) Write down a nonhomogeneous linear system with the trivial solution, or explain why this is impossible.
- (d) Write down a linear system with three equations that is equivalent to the system below.

$$\begin{aligned}x + 4y &= 12 \\ 3x + 8y &= 4\end{aligned}$$

- (e) **Turn-In!**

Find a and b so that the linear system below is consistent with infinitely many solutions.

$$\begin{aligned}ax + by &= 12 \\ 3x + 8y &= 4\end{aligned}$$

Solution.

- (a) One example is

$$\begin{aligned}x + y &= 1 \\ x + y &= 2\end{aligned}$$

which is inconsistent because the two equations contradict one another.

- (b) No such system exists. Every homogeneous linear system has the trivial solution, so it can never be inconsistent.
- (c) Solving the system gives the unique solution
- (d) No such system exists. The trivial solution cannot satisfy a nonhomogeneous system.
- (e) One equivalent system is obtained by adding a redundant equation, such as

$$\begin{aligned}x + 4y &= 12 \\ 3x + 8y &= 4 \\ 4x + 12y &= 16\end{aligned}$$

Worksheet Exercise 2 Reflect. Which of these questions ask for a specific example, and which are really asking you to think about what *must* be true of certain kinds of linear systems? What properties distinguish

homogeneous and nonhomogeneous systems?

Is it possible for a linear system to have exactly three solutions? Why or why not?

Solution. Homogeneous systems by definition are linear expressions (like $ax + by$, without a constant term) set equal to zero. You can tell if a system is homogeneous by checking if setting all the variables equal to zero is a solution. This is because setting all the variables equal to zero makes the left side zero, and the only way for there to be an equality is if the right side is also zero.

Linear systems have one, zero, or infinitely many solutions. This is a mathematical statement that we will prove later the course, but for right now, you can think about the way lines and planes can be arranged. A line can intersect another line at one point, or they can miss each other completely if they're parallel, or they can plot the exact same (infinitely many) points.

Worksheet Exercise 3 Computation Questions.. *Turn-In!*

For each linear system below, solve using the method of elimination to find its solutions or show that it has no solutions.

First system.

$$\begin{aligned}3x + 4y - z &= 8 \\6x + 8y - 2z &= 3\end{aligned}$$

Second system.

$$\begin{aligned}x + y &= 1 \\2x - y &= 5 \\3x + 4y &= 2\end{aligned}$$

Third system.

$$\begin{aligned}2x - y &= 5 \\4x - 2y &= t\end{aligned}$$

where t is any real number. For what value(s) of t is the system consistent? For what value(s) of t is the system inconsistent?

Solution. First system: If we try to eliminate x by multiplying the first equation by -2 and adding this to the second equation, we arrive at the false statement $0 = -13$. This system is therefore inconclusive. (Geometrically, these represent parallel planes in space!)

Second system: Eliminate x in the second equation using the first equation to arrive at the equivalent system

$$\begin{aligned}x + y &= 1 \\-3y &= 3 \\3x + 4y &= 5\end{aligned}$$

We can make an equivalent system by scaling the second equation to make our lives easier.

$$\begin{aligned}x + y &= 1 \\y &= -1 \\3x + 4y &= 5\end{aligned}$$

Now eliminate y from the first equation using the (new) second equation

$$\begin{aligned}x &= 2 \\y &= -1 \\3x + 4y &= 5\end{aligned}$$

We check to see if $(2, -1)$ is a solution to the final equation, but $3 \cdot 2 - 4 \cdot 1 = 2 \neq 5$, so the system is inconsistent.

Third system: if we try to eliminate variables, we end up with $2x - y = 5$ and $0 = t - 10$. For this system to be consistent, we must have $t = 10$, and then there are infinitely many solutions

$$\{(x, 2x - 5) : x \in \mathbb{R}\}$$

(i.e., the points on the line $y = 2x - 5$).